Leg Swelling and Chest Pain: Deep Vein Thrombosis Pulmonary Embolism Thrombophilia

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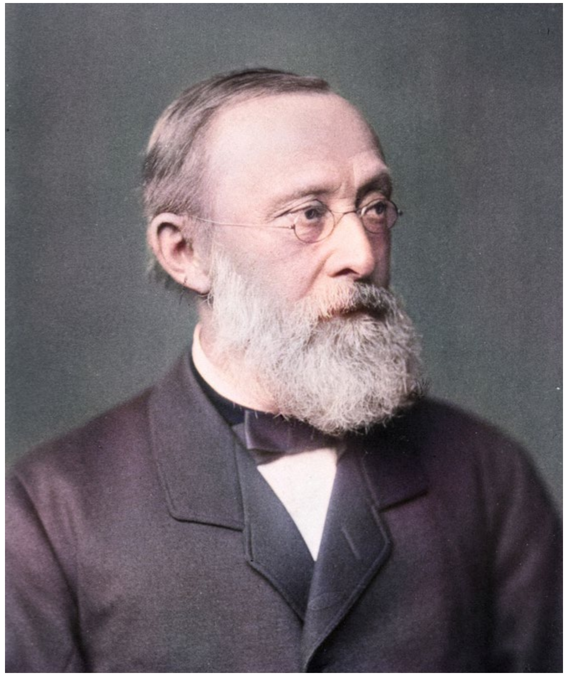
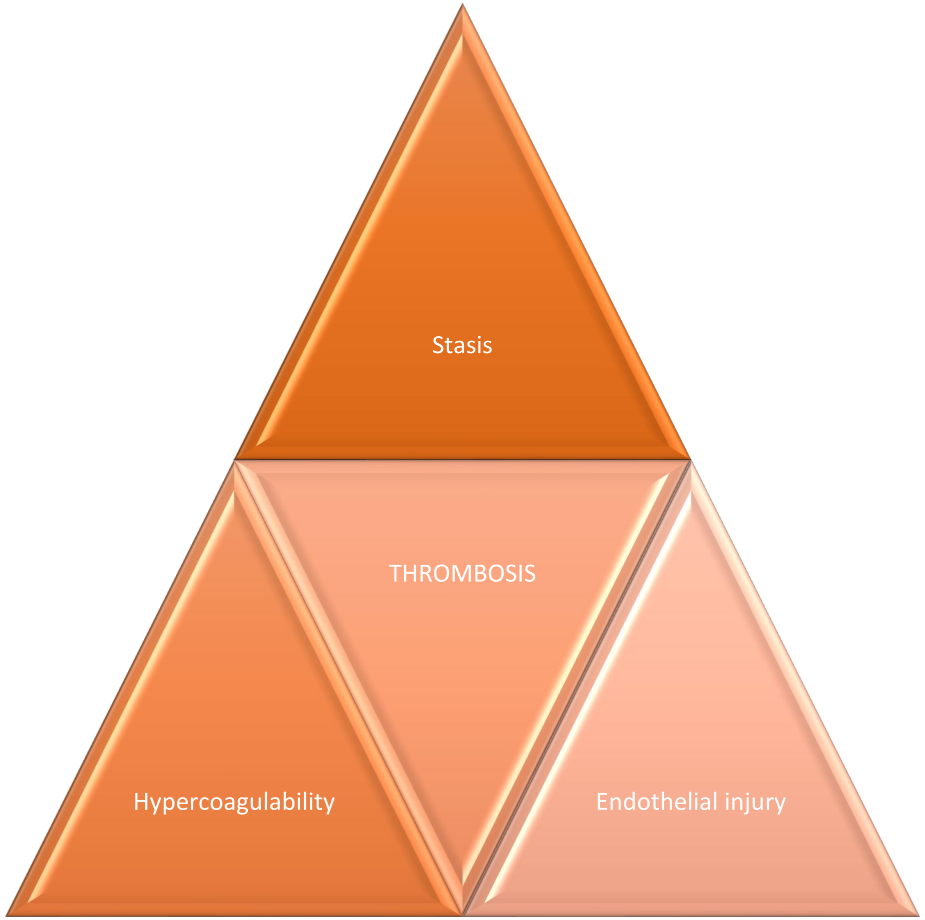
Lecture outline

- 1. Pathophysiology of venous thromboembolism
- 2. Risk factors of venous thromboembolism
- 3. Inherited and acquired thrombophilia
- 4. Clinical features of venous thromboembolism
- 5. Investigations for suspected venous thromboembolism
- 6. Treatment of venous thromboembolism

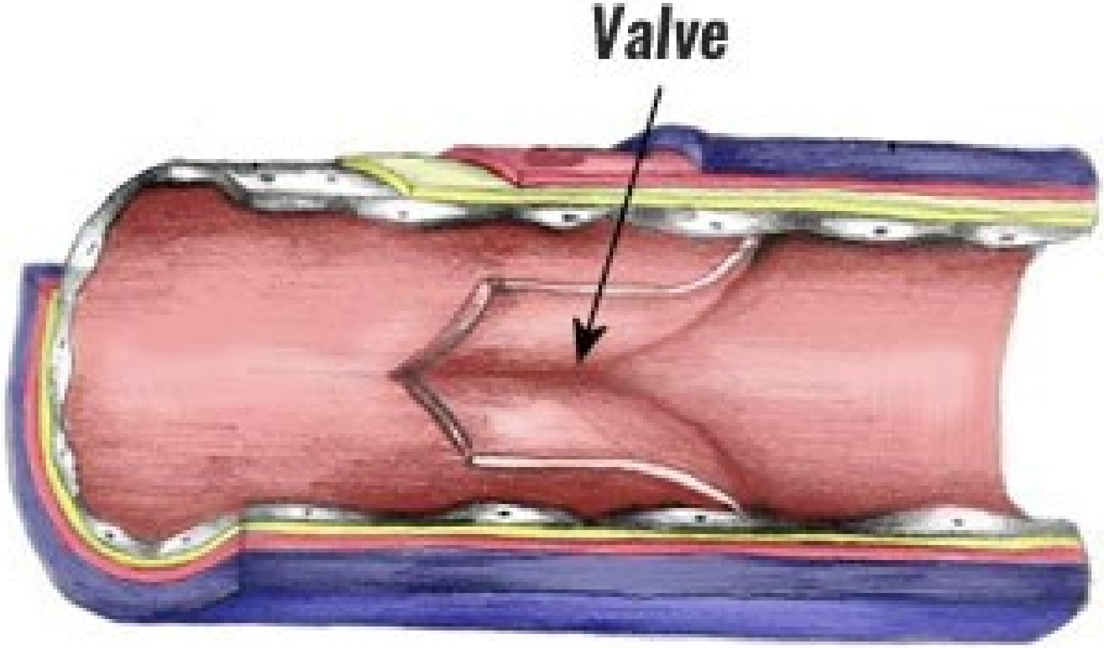
Virchow triad



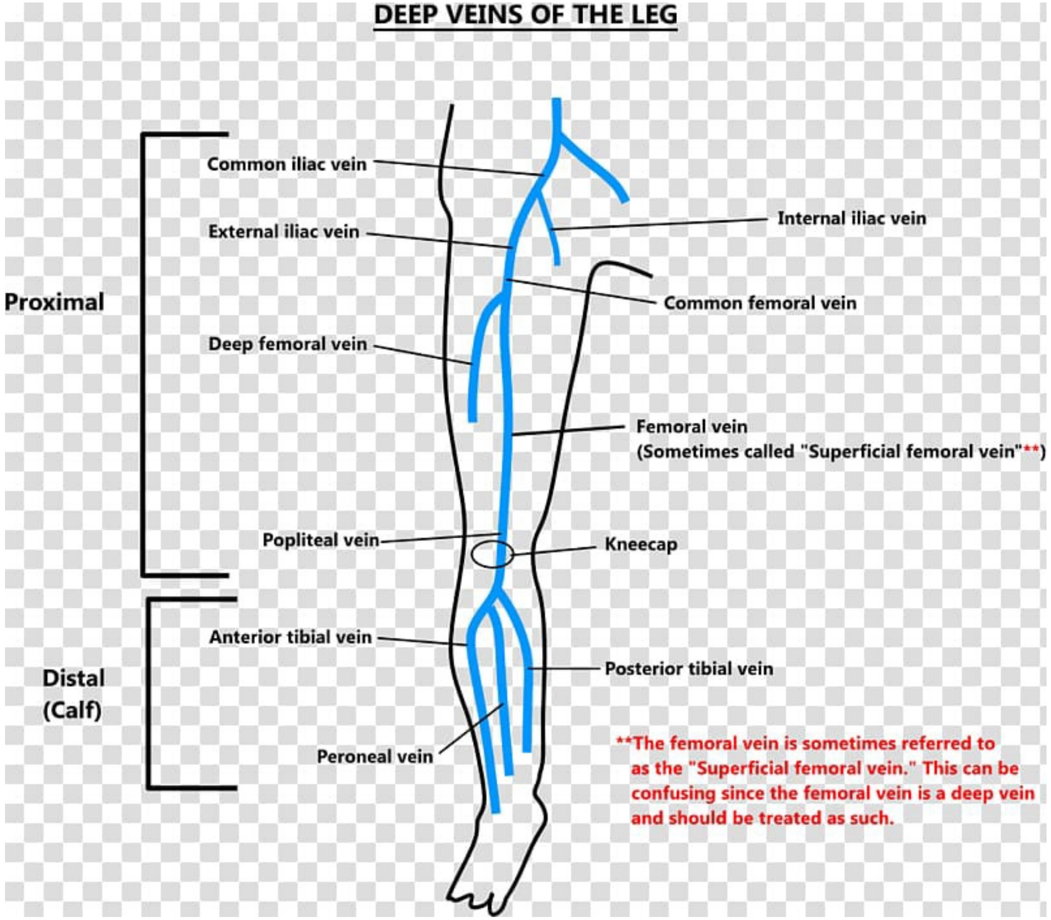


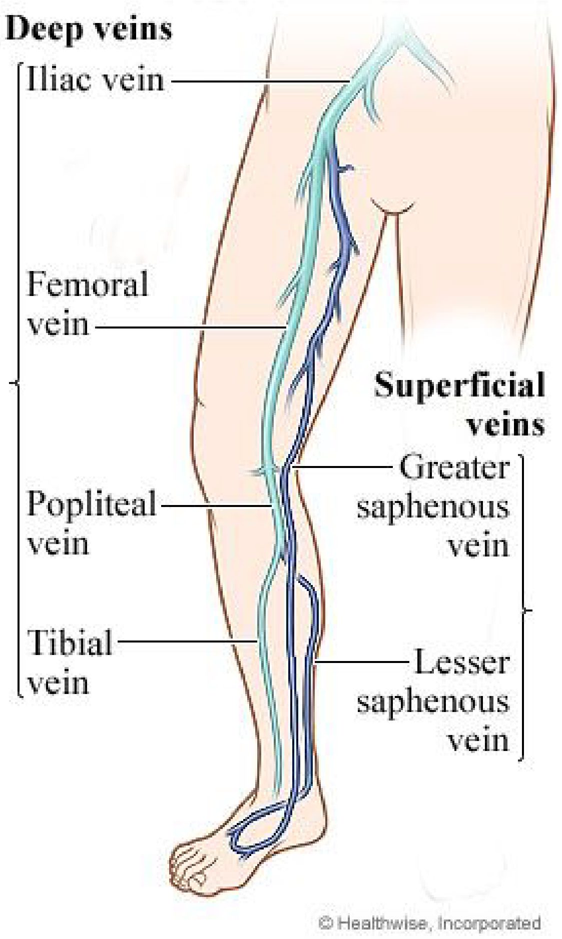
Venous thromboembolism

- Blood clot in VENOUS system
- Pulmonary embolism and deep vein thrombosis
- Low pressure system
- Superficial veins Vs deep veins



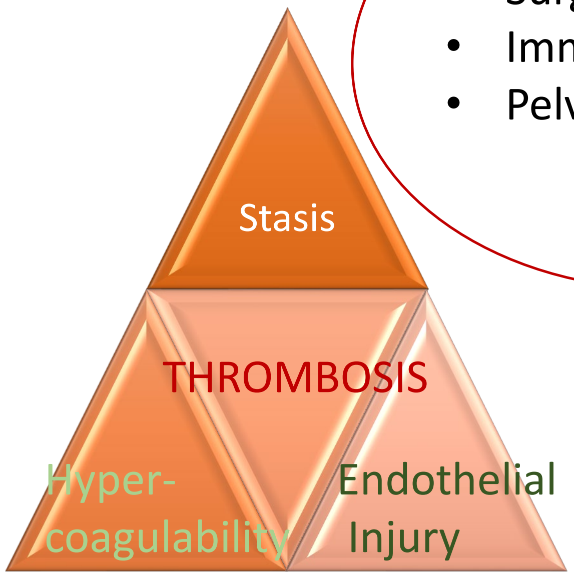
Superficial Vs Deep veins





Risk Factors of Venous thromboembolism

- Surgery
- Immobilization
- Pelvic obstruction

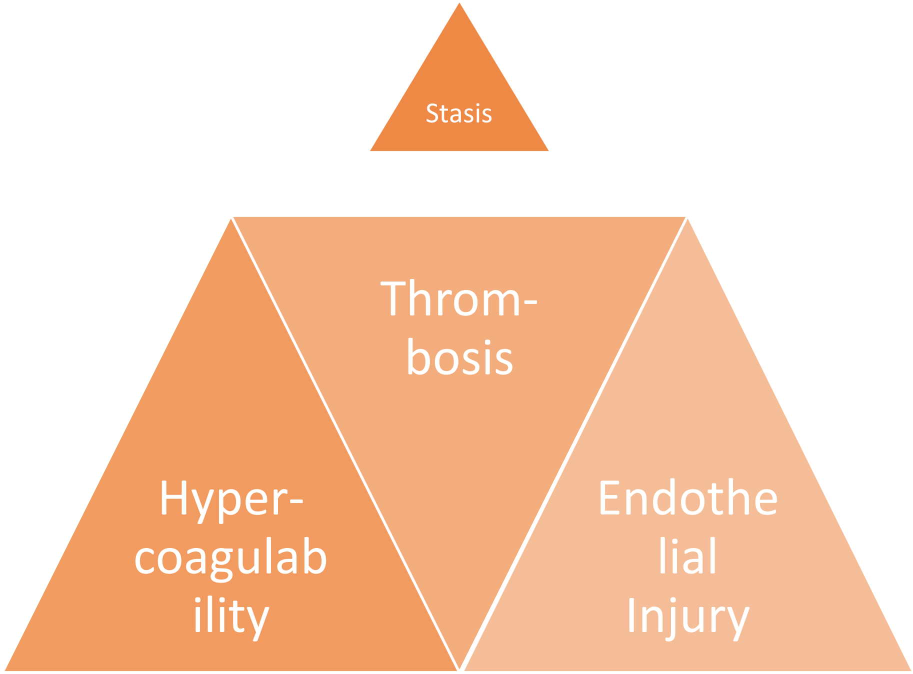


- Inherited thrombophilia
- Acquired thrombophilia
- Dehydration
- Hyperviscosity

- Surgery
- Radiation
- Indwelling catheter
- Sepsis

Economy Class Syndrome?

Risk factors of venous thromboembolism



Risk factors of venous thromboembolism

General	Stasis	Hypercoagulability	Endothelial injury
Age	Heart failure	Inherited thrombophilia - AT, PC, PS deficiency - Factor V Leiden - Hyperhomocysteinemia - Prothrombin G20210A mutation	Surgery
Obesity	Stroke	Acquired thrombophilia - Antiphospholipid syndrome	Radiation
Family history	Surgery	Malignancy	Tourniquet
	Pelvic obstruction	Nephrotic syndrome	Sepsis
	Varicose veins	Myeloproliferative neoplasm	
		Pregnancy or estrogen therapy	
		Dehydration	
		PNH	

Risk factors of arterial thrombosis

- Different spectrum of risk factors

Application to both arterial and venous	**Arterial thrombosis only**
Age	Hypercholesterolaemia
Family history	Diabetes Mellitus
Smoking	Hypertension
Obesity	Chronic renal impairment
Antiphospholipid syndrome	Male gender
Hyperhomocysteinaemia	
Myeloproliferative neoplasm	
PNH	

Inherited thrombophilia - antithrombin deficiency

- Antithrombin is synthesized in liver
- Antithrombin neutralizes thrombin, factor IXa, Xa, XIa, XIIa
- Activity of antithrombin is enhanced by heparin
- Autosomal dominant inheritance
- Estimated prevalence 1 in 2000 to 5000

Inherited thrombophilia – Protein C deficiency

- Protein C is synthesized in liver
- Vitamin K dependent anticoagulant, requiring activation by thrombin
- Inactivate factor Va and VIIIa
- Autosomal dominant inheritance
- Estimated prevalence 1 in 200 to 500

Inherited thrombophilia – Protein S deficiency

- Protein S is synthesized in liver, endothelial cells, megakaryocytes and brain cells
- Vitamin K dependent anticoagulant
- Protein S is cofactor of activated Protein C
- Autosomal dominant inheritance

Acquired deficiencies of natural anticoagulant

Antithr	
Neonat	
Pregna	
Liver d	
Sepsis	
Acute t	
DIC	
Nephr	
Hepari	
L-aspar	
Estrogen	Estrogen

NEVER DO THROMBOPHILIA SCREEN DURING ACUTE THROMBOSIS

Other inherited thrombophilia

- Factor V Leiden
  - Factor Va resistant to activated Protein C inactivation
  - Extremely rare in Asians
- Prothrombin G20210A mutation
  - Very rare in Asians
- Hyperhomocysteinemia
  - Inherited or acquired
  - Increased risk of atherosclerosis and venous thromboembolism
  - ?vascular injury, factor V activation

Antiphospholipid syndrome

- Antibodies against phospholipid binding proteins – lupus anticoagulant
  - Anti-cardiolipin antibodies (IgG or IgM)
  - Anti-B_2 glycoprotein I
  - Lupus anticoagulant

Persistence i.e. remains positive 12 weeks apart

- LA or ACA can be present as sole laboratory abnormality
  - Drugs
  - Infections
- Associated with recurrent arterial and venous thrombosis, livedo reticularis, recurrent fetal loss and thrombocytopenia

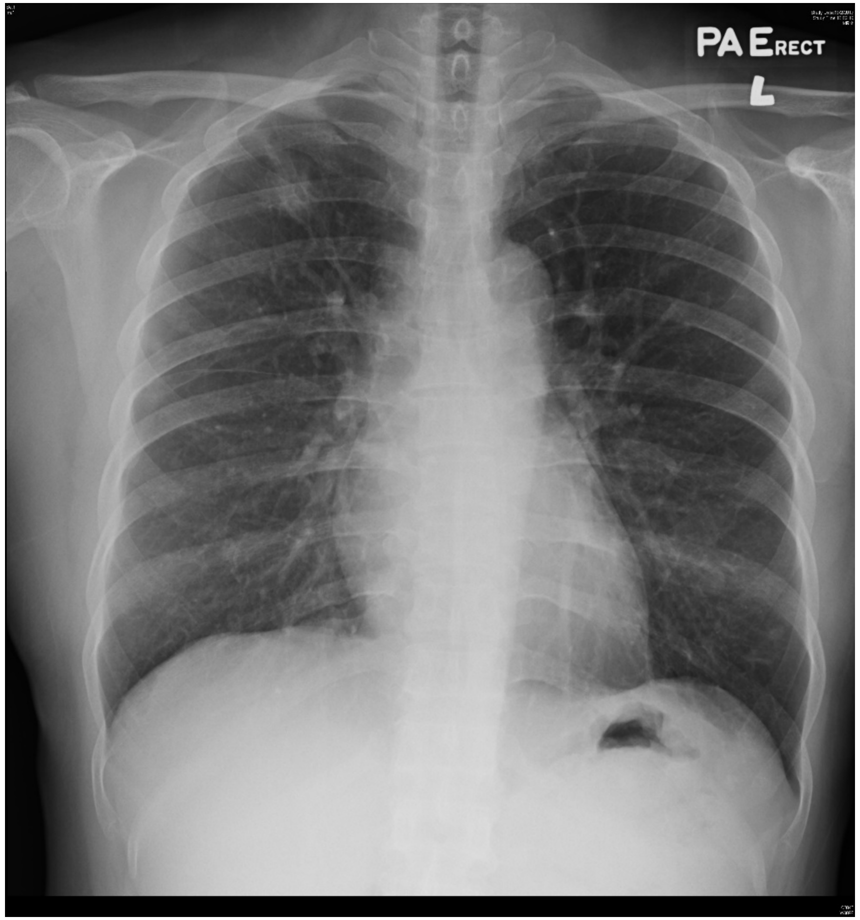
Antiphospholipid syndrome

- LMWH or warfarin in patients with APLS and VTE
- Anticoagulation should be indefinite
- Target INR is between 2 and 3
- DOAC is NOT recommended
- May consider to add aspirin if patients develop arterial thrombosis

Malignancy associated VTE

- Up to 10% of patients were found to have underlying malignancy after an "unprovoked" VTE
- Malignancy is a hypercoagulable state
- Thorough history, physical examination and appropriate investigations for every patient with unprovoked VTE
- Universal PET-CT screening is not recommended

A 52-year-old woman presented with LLL DVT



Myeloproliferative neoplasm associated thrombosis

- Myeloproliferative neoplasm increases both arterial and venous thrombotic risk
- JAK2 mutation increases thrombosis risk
- Particularly mesenteric thrombosis
- Beware of acquired von Willibrand disease in extreme thrombocytosis

Pregnancy and estrogen associated VTE

- Prevalence of VTE in pregnancy and puerperium 1 in 1600
- Highest risk during post-partum period
- Pregnancy is a hypercoagulable status
- Risk increases with
  - Multiple pregnancies
  - Maternal obesity
  - Maternal diabetes mellitus or hypertension
  - Known thrombophilia
  - Advanced maternal age
  - Hospitalization
  - Caesarean section
  - Eclampsia
  - Post-partum haemorrhage

Nephrotic syndrome and thrombophilia

- 8 times higher risk of having venous or arterial thrombosis in nephrotic syndrome patients
- Due to loss of natural anticoagulants through kidneys e.g. antithrombin
- Renal vein thrombosis is itself a cause of heavy proteinuria

Clinical features of venous thromboembolism

Clinical features of venous thromboembolism

- Most common presentation is lower limb deep vein thrombosis
  - Leg swelling, pain and/or warmth
  - Usually unilateral but could be bilateral
  - Ddx: cellulitis, ruptured Baker’s cyst, lymphedema, haematoma
- Pulmonary embolism
  - Pleuritic chest pain, hemoptysis, cough, dyspnea, syncope
  - Could be fatal
- Unusual site of thrombosis
  - Cerebral venous thrombosis
  - Mesenteric venous thrombosis

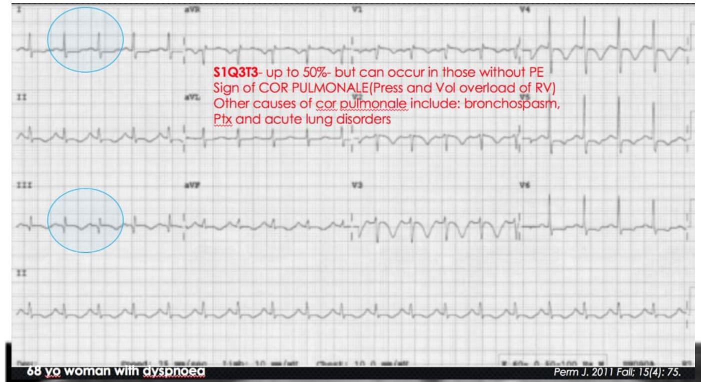
Approach to Thrombosis - History

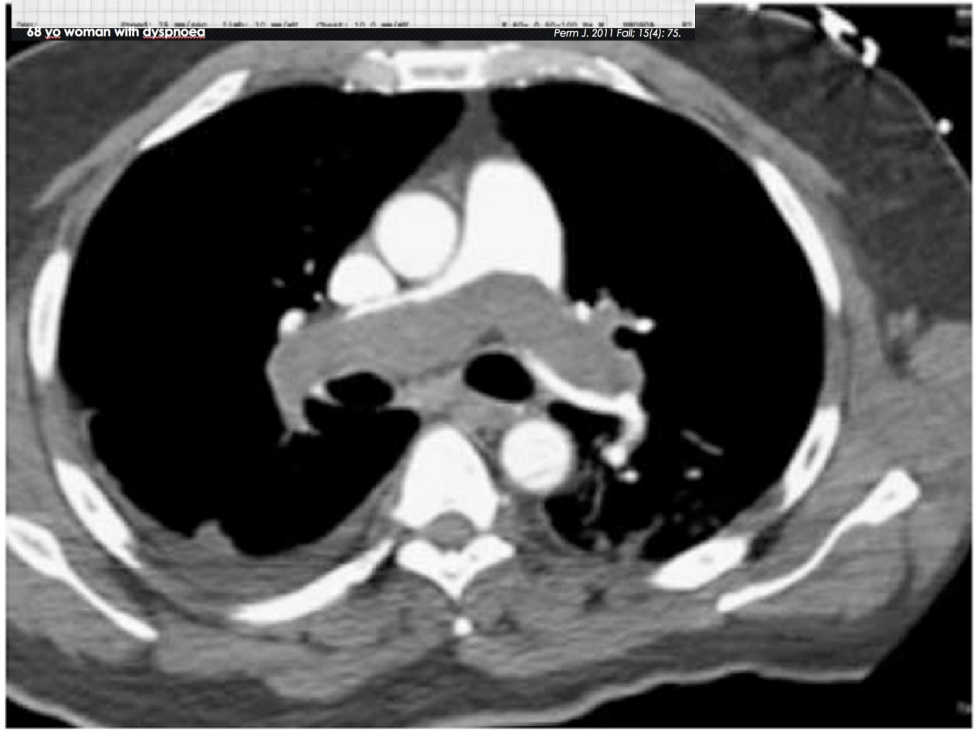
- Onset of symptoms – calf pain, swelling or chest symptom
- Screening of provocation
  - Recent operation
  - Trauma
  - Immobility
  - Pregnancy and obstetric history
  - OC pills or other hormones

- Family history of VTE
- Constitutional symptoms – THROMBOSIS CAN BE FIRST MANIFESTATION OF UNDERLYING MALIGNANCY

Investigations for suspected VTE

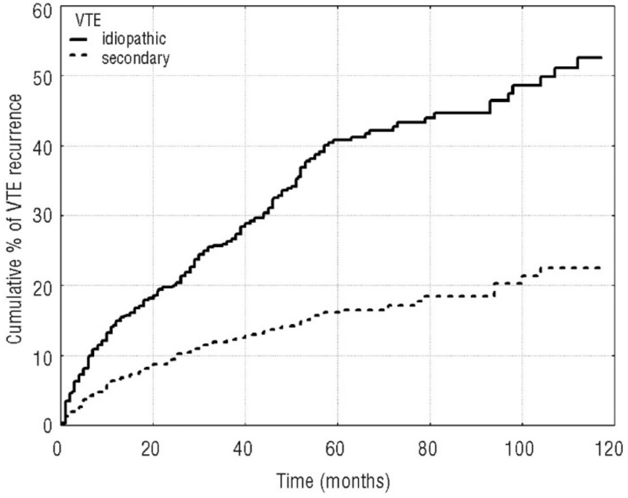
- Baseline CBP, clotting profile
- D-Dimer
- ECG
- Troponin
- USG Doppler of relevant body parts
- CT pulmonary angiogram
- Ventilation perfusion lung scan
- Echocardiogram





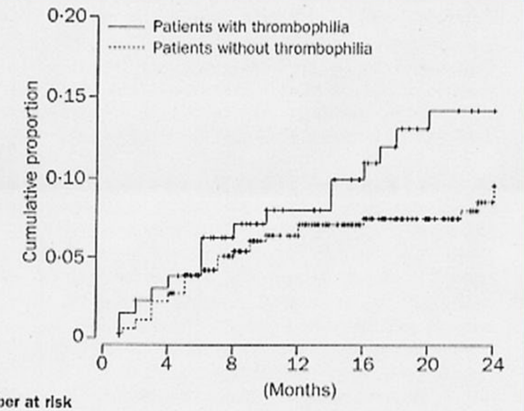
Thrombophilia screening

- Should not carry out thrombophilia screening during acute thrombosis
- Does not predict recurrence risk



	Idiopathic VTE			Secondary VTE		
months	N. of VTE	Cum Incid (%)	95% CI	N. of VTE	Cum Incid (%)	95% CI
6	85	10.0	8.0-12.0	32	4.2	2.8-8.7
12	39	15.0	12.6-17.4	17	6.6	4.8-8.4
36	74	26.3	23.0-29.6	34	12.3	9.8-14.8
60	56	40.8	36.5-45.1	14	16.1	13.0-19.2
96	9	46.4	41.1-51.8	6	20.3	15.8-24.8
120	5	52.6	45.6-59.5	2	22.5	17.2-27.8

From Baglin et al.
Lancet 2003



Number at risk

Patients with thrombophilia	130	125	111	100	90	76	71
Patients without thrombophilia	359	350	308	272	230	201	174

Cumulative proportions of recurrent thrombosis in patients with and without laboratory evidence of a heritable thrombophilic defect

Thrombophilia screening

- Not recommended following a provoked VTE
- Knowledge of inherited thrombophilia may allow
  - Provision of prophylaxis during at risk period e.g. pregnancy, operation
  - Identify family members at risk

- Screening of first degree relatives if anticoagulation cannot be discontinued in the index patient

Treatment

- Anticoagulation
  - LMWH
  - DOAC
  - Warfarin
- Thrombolytic
- Embolectomy
- IVC filter

Indication of thrombolytic therapy

- Thrombolytic therapy or embolectomy
- Hemodynamic instability – shock, persistent hypotension
- Venous gangrene of the limbs (Phlegmasia cerulea dolens)

Role of IVC filter

- Very limited role
- Efficacy based on limited uncontrolled case series
- May consider when
  - Toxicity of anticoagulation is predictably and unacceptably high
  - Temporary withhold anticoagulation yet preventing embolization of clot to pulmonary circulation
  - Plan for removal when the bleeding risk is lowered

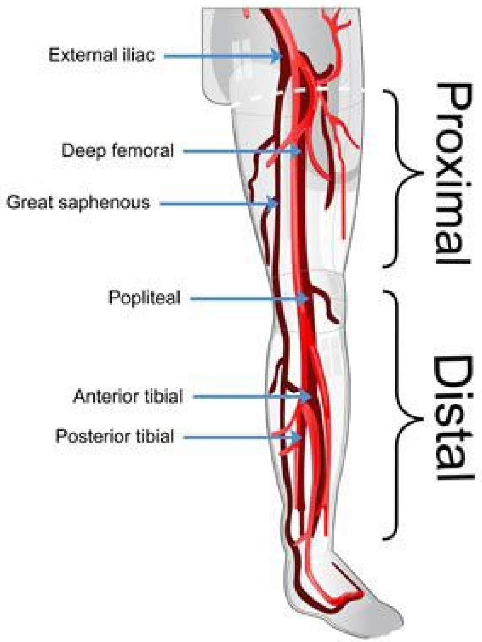
Duration of anticoagulation treatment

- Provoked event
  - 3 months
- Unprovoked event
  - 6 months or long term
- Recurrent event
  - Long term
- Underlying malignancy
  - As long as malignancy is active
- Underlying anti-phospholipid syndrome
  - Long term

Superficial venous thrombosis Vs Distal Vs Proximal DVT

- Superficial femoral vein is a DEEP vein of the thigh
- Superficial thrombophlebitis maybe manifestation of underlying malignancy
- Treatment: NSAIDS, LMWH or DOAC for 45 days

- Distal DVT carries a lower risk of embolization
- 3 months anticoagulation is adequate in most cases



Post-thrombotic syndrome

- Persistent swelling, pain, skin pigmentation and/or ulcers in the affected limb after DVT
- Due to chronic venous insufficiency
- Compression therapy

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The End